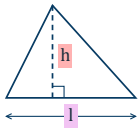
Nafn: _____

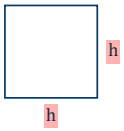
RÚMFRÆÐIFORMÚLUR

Þríhyrningur



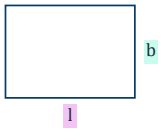
$$F = \frac{1}{2}hl$$

Ferningur



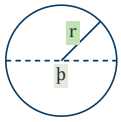
$$U = 4h$$
$$F = h^2$$

Rétthyrningur



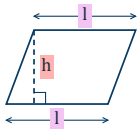
$$U = 2l + 2b$$
$$F = lb$$

Hringur



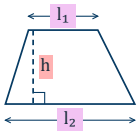
$$U = 2\,\pi\,r$$
$$\text{eða} U = \pi\,þ$$
$$F = \pi\,r^2$$

Samsíðungur



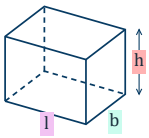
$$F = lh$$

Trapisa



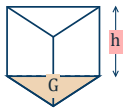
$$F = \frac{1}{2}h\left(l_1 + l_2\right)$$

Rétthyrndur strendingur



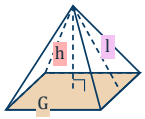
$$R = lbh$$
$$\text{YF} = 2lb + 2lh + 2bh$$

Strendingur



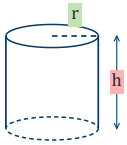
$$R = Gh$$
$$M = Uh$$
$$\text{YF} = Uh + 2G$$

Reglulegur píramídi



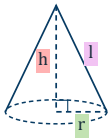
$$R = \frac{1}{3}Gh$$
$$M = \frac{1}{2}Ul$$
$$\text{YF} = \frac{1}{2}Ul + G$$

Sívalningur



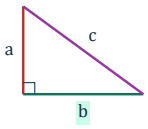
$$R = \pi\,r^2h$$
$$M = 2\,\pi\,rh$$
$$\text{YF} = 2\,\pi\,r^2 + 2\,\pi\,rh$$

Keila



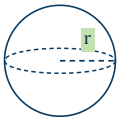
$$R = \frac{1}{3}h\,\pi\,r^2$$
$$M = \pi\,rl$$
$$\text{YF} = \pi\,r^2 + \pi\,rl$$

Regla Pýþagórasar



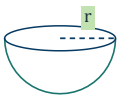
$$a^2 + b^2 = c^2$$

Kúla



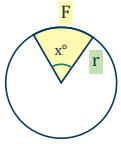
$$R = \frac{4}{3}\,\pi\,r^3$$
$$\text{YF} = 4\,\pi\,r^2$$

Hálfkúla með botni



$$R = \frac{2}{3}\,\pi\,r^3$$
$$\text{YF} = 3\,\pi\,r^2$$

Hringgeiri



$$F = \frac{x \cdot\,\pi\,r^2}{360}$$

Táknaskýringar

F	Flatarmál
YF	Yfirborðsflatarmál
R	Rúmmál
M	Möttulflötur
G	Grunnflötur
U	Ummál
l	Lengd
b	Breidd
h	Hæð / hlið
r	Radíus
$þ$	Þvermál

G: flatarmál grunnflatar. **M:** flatarmál möttuls. **U:** ummál grunnflatar í rúmmyndum.
Strendingar og sívalningur eru réttir; keilan er rétt og píramídinn reglulegur. Hálfkúlan er með botni. **l** táknar hæð hliðaflatar píramída og skáhæð keilu.

Fyrirmynd: © Gina Wilson (All Things Algebra), 2015. Íslensk aðlögun og endurteikning.

Nafn: _____

RÚMFRÆÐI

Talnahús fyrir mælieiningar

Metrar

Stórar einingar			Hefðbundnar mælieiningar							Litlar einingar		
10^{12}	10^{9}	10^{6}	10^{3}	10^{2}	10^{1}	$10^{0} = 1$	10^{-1}	10^{-2}	10^{-3}	10^{-6}	10^{-9}	10^{-12}
Tm	Gm	Mm	km	hm	dam	m	dm	cm	mm	µm	nm	pm
Tera-	Gíga-	Mega-	Kíló-	Hektó-	Deka-	Metri	Desi-	Senti-	Milli-	Míkró-	Nanó-	Píkó-

Fermetrar

km^2		hm^2		dam^2		m^2		dm^2		cm^2		mm^2	
	km^2		hm^2		dam^2		m^2		dm^2		cm^2		mm^2

Rúmmetrar

km^3			hm^3			dam^3			m^3			dm^3			cm^3			mm^3		
		km^3			hm^3			dam^3			m^3			dm^3			cm^3			mm^3
														l	dl	cl	ml			